

CHAPTER 17 SUMMARY

Current and Resistance

(Section 17.1) Electric current flows from higher to lower electric potential. An ohmic material's resistance is constant, independent of potential difference, although it varies with temperature.

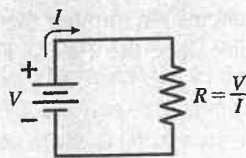
Electrical resistivity is a property of a material; resistance of a piece of material depends on size and shape as well as resistivity.

$$\text{Electric current: } I = \frac{\Delta q}{\Delta t}$$

$$\text{Resistance: } R = \frac{V}{I}$$

Temperature dependence of resistivity or resistance:

$$\rho = \rho_0[1 + \alpha(T - T_0)]$$



Batteries: Real and Ideal

(Section 17.2) Batteries have a fixed potential difference (emf) and supply current in closed circuits.

Kirchhoff's loop rule says that the sum of the potential differences around a closed loop is zero.

Combining Resistors

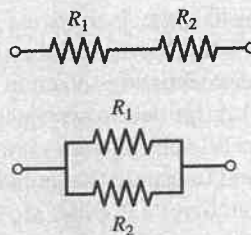
(Section 17.3) For resistors in series, the equivalent resistance is the sum of all the resistances.

Equivalent resistance of resistors in series: $R_s = R_1 + R_2$.

Kirchhoff's junction rule says that the net current entering a junction equals the net current leaving the junction. Resistors in parallel add reciprocally.

Reciprocal of equivalent resistance for resistors in parallel:

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$$

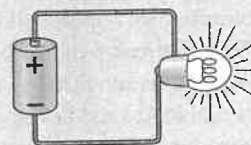


Electric Energy and Power

(Section 17.4) Resistors dissipate electrical energy.

The rate of energy dissipation (power) can be expressed in terms of any two of three quantities: potential difference, current, and resistance.

$$\text{Electric power with resistor in a circuit: } P = IV = I^2R = \frac{V^2}{R}$$



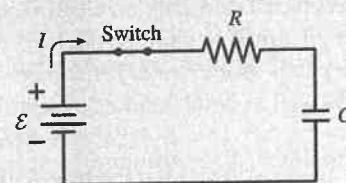
RC Circuits

(Section 17.5) In an RC circuit, the capacitor charges gradually, with time constant RC.

Charge and potential difference decay exponentially in a discharging capacitor, with time constant RC.

$$\text{RC circuit charging: } q(t) = \mathcal{E}C(1 - e^{-t/RC})$$

$$\text{Discharging: } q(t) = q_0 e^{-t/RC}$$



NOTE: Problem difficulty is labeled as ■ straightforward to ■■■ challenging. Problems labeled BIO are of biological or medical interest.

Conceptual Questions

1. If resistance is due to collisions between conduction electrons and ions in a solid, why should resistance increase approximately linearly with temperature? *Hint:* Think about the oscillations of the ions in the solid.
2. Explain why the electron drift speed (Equation 17.2) is proportional to I/A .
3. Is thermal expansion (Chapter 12) an important factor in how a metal conductor's resistance changes as a function of temperature? Explain.
4. Which has the higher resistance, a 60-W lightbulb or a 100-W lightbulb? Each is designed to operate at 120 V.
5. How can $P = I^2R$ and $P = V^2/R$ both be true?
6. Express the ohm (Ω) in terms of kg, m, s, and C.

In Conceptual Questions 7–9, you have a battery and a box of identical lightbulbs. Let a single bulb attached to the battery serve as a reference (Figure CQ17.7).

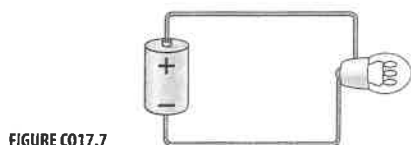


FIGURE CQ17.7

7. Three bulbs are connected in series across the battery. How do the bulbs' brightnesses compare with each other and with the reference bulb? What happens to the relative brightness if one bulb is unscrewed?
8. Three bulbs are connected in parallel across the battery. How do their brightnesses compare with each other and with the reference bulb? What happens to the relative brightness if one bulb is unscrewed?
9. Three bulbs are connected as shown (Figure CQ17.9), with a parallel connection of two bulbs in series with a third. How do their brightnesses compare with each other and with the reference bulb? What happens to the relative brightness if B_1 is removed while the others are left in place? Repeat for the case when B_2 alone is removed.

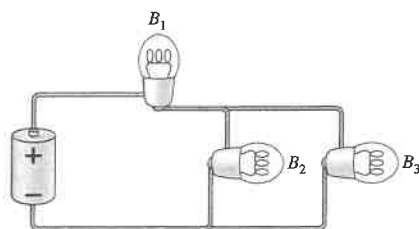


FIGURE CQ17.9

10. Show that the units on the right-hand side of $P = I^2R$ reduce to J/s or W.
11. Is the power consumed by a lightbulb greater immediately after it's turned on (when the filament is still cool) or later, after the filament has warmed?
12. Explain why the energy loss in power lines carrying a given power is lower if a higher potential is maintained between the wires.

13. Two cylindrical wires are made of the same material, but one has a larger diameter. Which wire is more likely to melt if they're each connected across the same battery?
14. If you double the length of the heating wire in your toaster, will you get more heating power, less, or the same amount? Explain.
15. Why is it safe for a bird to sit on a power line? Could you safely sit on the line next to the bird?
16. What's the potential difference between two ends of a superconducting wire carrying a current I ? How does $I = V/R$ make sense here?
17. Are the electrical outlets in your home connected in series or parallel? How do you know?
18. Would a superconducting wire make a good lightbulb filament? Why or why not?

Multiple-Choice Problems

19. The resistance of a 100-m-long copper wire with diameter 2.0 mm is (a) 540 Ω ; (b) 0.54 Ω ; (c) 1.9 Ω ; (d) 190 Ω .
20. The electron drift speed in a 0.75-mm-diameter copper wire carrying a current of 1.2 A is (a) 0.2 mm/s; (b) 2.0 mm/s; (c) 0.05 mm/s; (d) 5.0 mm/s.
21. What resistor would have a current of 1.5 mA when connected across a 1.5-V battery? (a) 10 Ω ; (b) 15 Ω ; (c) 1 k Ω ; (d) 1.5 k Ω .
22. An 18.1-V battery is attached to a 25.4- Ω load, and the resulting current is 678 mA. The battery's internal resistance is (a) 0.013 Ω ; (b) 0.77 Ω ; (c) 1.3 Ω ; (d) 1.7 Ω .
23. A 1.51-V battery with internal resistance 1.25 Ω delivers 0.125 A to its load. The load resistance is (a) 5.4 Ω ; (b) 7.9 Ω ; (c) 10.8 Ω ; (d) 54 Ω .
24. The equivalent resistance of a 120-k Ω resistor and a 140-k Ω resistor in parallel is (a) 260 k Ω ; (b) 20 k Ω ; (c) 0.15 k Ω ; (d) 65 k Ω .
25. For any two parallel resistors, (a) the current in each resistor is the same; (b) the power dissipated by each resistor is the same; (c) the potential difference across each resistor is the same; (d) none of the above is necessarily true.
26. A 320- Ω resistor and a 530- Ω resistor are in series. When this combination is connected across a 12-V battery, the current is (a) 14 mA; (b) 23 mA; (c) 38 mA; (d) 140 mA.
27. You have a 240- Ω resistor and wish to combine it with a second resistor to make an equivalent resistance of 200 Ω . You should (a) add a 40- Ω resistor in series; (b) add a 40- Ω resistor in parallel; (c) add a 440- Ω resistor in parallel; (d) add a 1200- Ω resistor in parallel.
28. A 9.0-V battery, 100- Ω resistor, and 240- Ω resistor are in series. The power delivered by the battery is (a) 0.24 W; (b) 4.2 W; (c) 0.026 W; (d) 0.09 W.
29. In an RC circuit with $R = 350$ k Ω and $C = 200$ μ F, the time constant is (a) 1.75 s; (b) 70 s; (c) 700 s; (d) 7000 s.
30. The resistor you should put in series with a 250-pF capacitor to make the time constant 1 ms is (a) 250 k Ω ; (b) 2.5 M Ω ; (c) 4.0 M Ω ; (d) 7.5 M Ω .
31. If you discharge a 60- μ F capacitor through a 1.5-k Ω resistor, how much time does it take for 99.9% of the charge to leave the capacitor? (a) 0.62 s; (b) 0.09 s; (c) 1.75 s; (d) 150 s.

Problems

Section 17.1 Current and Resistance

32. ■ In a 0.50-mm-diameter copper wire, what current corresponds to an electron drift speed of 1.0 m/s? Do you think this is practical?
33. ■ A thin wire made of tin with diameter 2.5×10^{-5} m carries 0.15 mA of current. Find the electron drift speed.
34. ■ What diameter should you make a copper wire so its resistance is 0.010Ω per meter of length?
35. ■ Find the resistance between the two ends of an aluminum wire 0.25 m long and 1.0 cm in diameter.
36. ■■ A platinum wire is originally at 20°C . Find the temperature required to raise its resistance by (a) 1% and (b) 10%.
37. ■ In a lightning strike lasting 45 ms, a total charge of 350 C is transferred to the ground. What's the average current?
38. ■■ An electron beam carries 1.0 mA. How many electrons pass a given point each second?
39. ■ Find the resistance of a 1.0-m length of silver wire with diameter (a) 5.0×10^{-5} m; (b) 5.0×10^{-4} m; (c) 5×10^{-3} m.
40. ■■ You have some copper wire that you want to connect from a power supply to a $150\text{-}\Omega$ load. The wire is a standard 24-gauge (diameter 0.51 mm). What's the maximum length of wire you can use without its resistance exceeding 1% of the load resistance? *Note:* You're finding the total length of wire to and from the load.
41. ■■ A platinum wire with room-temperature resistance 58.5Ω is used as a thermistor. If you want to measure the temperature to within 0.1°C in a range around room temperature, how precisely should the wire's resistance be measured?
42. ■■■ A nichrome toaster wire, initially at room temperature, carries 2.75 A immediately after it's turned on. After the wire heats, the current drops to 2.46 A. What's the temperature of the nichrome then?
43. ■■■ Two cylindrical metal wires, A and B, are made of the same material and have the same mass. Wire A is twice as long as wire B. What's the ratio R_A/R_B ?
44. ■■ A 2.50-m-long copper wire has 1.50 V across its ends. (a) Find the electric field strength within the wire. (b) What's the acceleration of an electron in this field? (c) A typical drift speed for conduction electrons is 1 mm/s. How far would an electron travel before reaching this speed?

Section 17.2 Batteries: Real and Ideal

45. ■■ (a) For how much time must a 12.0-V battery be connected across a $1.50\text{-k}\Omega$ load to move 1.0 C of charge through the load? (b) How much energy does the battery provide during that time?
46. ■■ A 12.0-V battery produces 230 mA when it's across a $50.0\text{-}\Omega$ load. (a) What's the battery's internal resistance? (b) Find the potential difference across the battery's terminals.
47. ■■ When an 18.0-V battery is connected across a $130\text{-}\Omega$ load, the potential difference between the battery terminals is 17.4 V. What's the internal resistance?
48. ■■ A 1.51-V battery has internal resistance 1.35Ω . The battery powers a lightbulb whose filament (when lit) has resistance 8.55Ω . Find (a) the current through the lightbulb and (b) the potential difference across the battery terminals with the bulb connected.
49. ■■ The nickel-metal hydride battery in your cell phone has a 1.2-V emf, and it can be recharged about 1000 times. Manufacturers often rate batteries in milliamphours (mA-h) they can supply between rechargings. (a) Show that the mA-h is a charge unit, and find the conversion factor from mA-h to C. (b) How much charge can a 1400-mA-h battery supply? (c) How much total energy can that battery supply between rechargings?
50. ■■ In a rechargeable battery, the charging process consists of reversing the normal current flow, forcing current into the positive battery terminal. Suppose it takes 3.2×10^5 C to recharge a 9-V battery. (a) How much work must be done to move that much charge against the battery's 9-V emf? (b) If the charging process takes 2.5 h, what's the average power required?
51. ■■ A battery produces 15.5 mA when it's connected to a $230\text{-}\Omega$ load and 22.2 mA when it's connected to a $160\text{-}\Omega$ load. Find the battery's emf and internal resistance.

Section 17.3 Combining Resistors

52. ■ A $34\text{-}\Omega$ and a $45\text{-}\Omega$ resistor are in series. (a) Find the equivalent resistance. (b) Find the current when this series combination is connected across a 7.5-V battery.
53. ■ Two resistors, 250Ω and 370Ω , are in parallel. (a) Find the equivalent resistance. (b) The parallel combination is connected across a 12-V battery. Find the current supplied by the battery and the current in each resistor.
54. ■■ You have three $100\text{-}\Omega$ resistors. List all the possible equivalent resistance values you can make using one, two, or three resistors in combination.
55. ■■ Four resistors, 12Ω , 15Ω , 20Ω , and 35Ω , are in series. (a) Find the equivalent resistance. (b) Find the circuit current and the potential difference across each resistor when the series combination is across a 6.0-V battery. Verify that the sum of the four potential differences equals the battery emf.
56. ■■ The resistors of the preceding problem are now connected in parallel. (a) Find the equivalent resistance. (b) Find the current in each resistor when the parallel combination is across a 6.0-V battery. Verify that the sum of the resistor currents equals the battery current.
57. ■■ You're given a box of $10\text{-}\Omega$ resistors. How would you make the following equivalent resistances: (a) 2Ω ; (b) 35Ω ; (c) 7Ω ; (d) 19Ω ?

Problems 58–61 refer to the accompanying diagram (Figure P17.58).

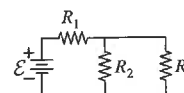


FIGURE P17.58

58. ■■ If $\mathcal{E} = 9.0$ V, $R_1 = 150 \Omega$, $R_2 = 250 \Omega$, and $R_3 = 1000 \Omega$, find the current in each resistor.
59. ■■ Assume all the resistors are identical. If the battery's emf is 12 V and it supplies 200 mA, find each resistance.
60. ■■ Suppose $\mathcal{E} = 10.0$ V, $R_1 = 200 \Omega$, and $R_2 = 300 \Omega$. What value of R_3 would you select so the battery supplies 25.0 mA?
61. ■■ The three resistors each have resistance $50 \text{ k}\Omega$. What should be the battery's emf if it's to supply 1.0 mA of current?
62. ■■ A $1.5\text{-M}\Omega$ resistor and a $2.5\text{-M}\Omega$ resistor are in series with a battery. The potential difference across the $1.5\text{-M}\Omega$ resistor is measured at 6.0 V. What is the emf of the battery? (b) What is the circuit current?
63. ■■ A $20\text{-k}\Omega$ resistor, a $30\text{-k}\Omega$ resistor, and a $75\text{-k}\Omega$ resistor are in parallel across a battery. The current in the $20\text{-k}\Omega$ resistor is 0.125 mA. Find (a) the battery emf and (b) the current in each of the other resistors.
64. ■■ A series combination of four identical resistors is across a battery. The potential difference across each resistor is 2.2 V, and

the circuit current is 0.26 A. Find the value of each resistor and the battery emf.

Section 17.4 Electric Energy and Power

65. ■ A $150\text{-}\Omega$ resistor is across a 24.0-V battery. Find the energy supplied by the battery in 1 min.
66. ■ What is the resistance (when lit) of a lightbulb rated at 120 V and 100 W?
67. ■■ Carbon resistors commonly come in ratings of $1/8\text{ W}$, $1/4\text{ W}$, $1/2\text{ W}$, or 1 W , meaning that this is the maximum rate at which the resistor can dissipate energy at room temperature. If a $280\text{-}\Omega$ resistor has a rating of $1/2\text{ W}$, what's the maximum current it should carry?
68. ■■ (a) A battery supplies 100 W with a $20\text{-}\Omega$ resistor across its terminals. Find (a) the battery emf and (b) the resistor current.
69. ■■ A hair dryer consumes 1750 W when connected to a 120-V source. (a) Find the current in the hair dryer. (b) How much energy is consumed if the dryer runs for 10 min? (c) Compute the monthly cost if the hair dryer runs for 10 min each day at a cost of 15¢ per kilowatt-hour.
70. ■■ A 0.10-mm-diameter copper wire is 15 m long. (a) Find its resistance. (b) What's the power dissipation in the wire if the potential difference across its ends is 12 V? (c) Repeat parts (a) and (b) if the wire is immersed in liquid nitrogen at 77 K.
71. ■■ A group of 200-W floodlights is wired in parallel to a 120-V electrical outlet to light an outdoor basketball court. What's the maximum number of lights that can be used without tripping the 20-A circuit breaker?
72. ■■■ A tungsten lightbulb filament is designed to dissipate 100 W at a 120-V potential difference. (a) What's the filament current, assuming negligible resistance in the rest of the circuit? (b) What is the filament's resistance when lit? (c) If the lit filament is at 1850 K, what's its resistance when off?
73. ■■■ An electric space heater draws 7.25 A at 120 V. (a) At what rate does it supply heat? (b) If the heater's energy goes into warming the air in an empty room (3.0 m by 3.0 m by 2.5 m), by how much does the air temperature rise in 1 min? (The specific heat capacity of air is about $1.0\text{ kJ}/(\text{kg} \cdot \text{K})$.)

Section 17.5 RC Circuits

74. ■ How large a capacitor should you combine with a $450\text{-k}\Omega$ resistor to make an RC time constant of 25 ms?
75. ■■ A 300-pF capacitor, initially uncharged, is in series with a $175\text{-M}\Omega$ resistor. The series combination is then connected across a 6.0-V battery. Find the charge on the capacitor after (a) 1.0 ms; (b) 10 ms; (c) 100 ms.
76. ■■ Find the number of time constants for the capacitor in an RC charging circuit to reach 99% of its maximum charge.
77. ■■■ How many time constants does it take for a charging capacitor to reach (a) half of its maximum charge? (b) half its maximum stored energy?
78. ■■ A 25-nF capacitor is in series with a $1.45\text{-M}\Omega$ resistor. The capacitor is initially uncharged when the series combination is connected across a 24.0-V battery. (a) What's the maximum charge the capacitor will eventually attain? (b) How much time does it take for the capacitor to reach 90% of that maximum charge?
79. ■■ A series RC circuit consists of a $550\text{-}\mu\text{F}$ capacitor (initially uncharged) and a $12\text{-k}\Omega$ resistor. The combination is connected across a 12-V battery; 3.0 s later, what's (a) the charge on the capacitor and (b) the current in the resistor?

80. ■■ A $150\text{-}\mu\text{F}$ capacitor charged initially to 0.28 mC is connected across a resistor, and 2.0 s later its charge has dropped to 0.21 mC. Find the resistance.

81. B10 ■■ **Defibrillator.** A defibrillator contains a $210\text{-}\mu\text{F}$ capacitor, charged by a 2.4-kV source through a $31\text{-k}\Omega$ resistance. (a) What's the capacitor charge after 20 s? (b) What fraction of the maximum possible charge is your answer to part (a)? (c) How much charge would be added if the charging continued for another 20 s?

Section 17.6 Semiconductors and Superconductors

82. ■■ A section of copper transmission line carries a steady current of 250 A and has resistance $0.76\text{ }\Omega$. Suppose this line is replaced by a superconducting line. (a) Ignoring the energy needed to refrigerate the superconductor, how much energy will be saved in one year? (b) What's the cost of that energy, if the utility company's price is 11¢ per kilowatt-hour?

General Problems

83. ■■ A $2.5\text{-k}\Omega$ resistor and 4.4-mF capacitor are in series, and the combination is connected across a 15-V power supply. Find the charge on the capacitor and current in the resistor after a time of (a) 1.0 s and (b) 10 s.
84. ■■■ You have four $100\text{-}\Omega$ resistors, each capable of dissipating $1/2\text{ W}$. How could you combine them to make a $100\text{-}\Omega$ resistor capable of dissipating 2 W?
85. ■■ You are building a circuit and need a $3.44\text{-k}\Omega$ resistor. You have three resistors: $1.50\text{-k}\Omega$, $3.30\text{-k}\Omega$, and $4.70\text{-k}\Omega$. How can you connect them to get the required resistance?
86. ■■ You're about to purchase a battery. Normally, batteries are rated in ampere-hours ($\text{A} \cdot \text{h}$)—the total amount of charge they can deliver. The 6-V battery you see is rated at 50 watt-hours, meaning the total amount of energy it can deliver, but your application calls for a $5.0\text{-A} \cdot \text{h}$ battery. Will this battery work?
87. ■■ A $50\text{-}\Omega$ resistor is connected across a battery, and a 26-mA current flows. When the $50\text{-}\Omega$ resistor is replaced with a $22\text{-}\Omega$ resistor, a 43-mA current flows. What are the battery's emf and internal resistance?
88. ■■ Find the resistance needed in an RC circuit to bring a $20\text{-}\mu\text{F}$ capacitor from zero charge to 45% of its full charge in 140 ms.
89. ■■ A capacitor is charged until it holds 5.0 J of energy. It is then discharged by connecting it across a $10\text{-k}\Omega$ resistor. In the first 8.6 ms after making the connection, the resistor dissipates 2.0 J. What is the capacitance?
90. B10 ■■ **Ion channel.** The potential difference across the ion channel described in Example 17.1 is 80 mV. (a) What's the resistance of the ion channel? (b) The ion channel can be modeled as a cylinder with a length of 6.0 nm and a diameter of 0.35 nm. What's the resistivity of the fluid filling the channel? (c) At what rate is energy dissipated while the ions are flowing?
91. ■■■ A camera's flash uses the energy stored in a $240\text{-}\mu\text{F}$ capacitor charged to 170 V. The current in the charging circuit is not to exceed 25 mA. (a) What's the resistance in the charging circuit? (b) How long do you have to wait between taking flash pictures? Consider that the capacitor is essentially at full charge in 5 time constants.
92. ■■■ Consider the circuit in Figure GP17.92. (a) Find the equivalent resistance of the entire network. (b) Find the current through each resistor and the potential difference across each resistor.

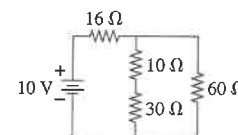


FIGURE GP17.92

93. ■■■ For the circuit in Figure GP17.93, find (a) the equivalent resistance of the network, (b) the current through each resistor, and (c) the power supplied by the battery.

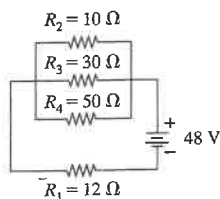


FIGURE GP17.93

94. ■■■ Repeat the preceding problem for the circuit in Figure GP17.94.

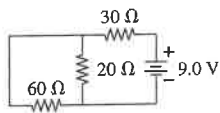


FIGURE GP17.94

95. B10 ■■ **Electric eel.** An electric eel uses a number of small cells called electrocytes, placed end to end in series, to produce a large potential difference. (a) If the eel produces a potential difference of 600 V, how many electrocytes, each with a potential difference of 125 mV, are required? (b) The eel can produce electrical energy at a rate of 600 W for a short time. What's the current required to do this?
96. ■■■ Figure GP17.96 shows the potential difference across a capacitor that's charging through a 4700-Ω resistor in the circuit of

Figure 17.20. Use the graph to determine (a) the battery's emf; (b) the capacitance; (c) the total energy dissipated by the resistor during the 10 ms interval shown.

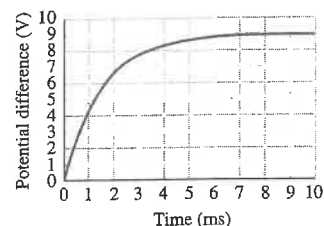


FIGURE GP17.96

Answers to Chapter Questions

Answer to Chapter-Opening Question

The apparatus measures the body's electrical resistance by passing a small current between electrodes attached to the skin. Fat has a higher resistance than lean tissue, so the result gives an indication of the body's fat content.

Answers to GOT IT? Questions

Section 17.1 (b) > (a) = (d) > (c)

Section 17.2 (d) 4 Ω

Section 17.3 (a) > (c) > (d) > (b)

Section 17.4 (a) 0.5 A, 240 Ω

Section 17.5 (d) > (b) > (a) > (c)